Air Quality Index Analysis: Madhya Pradesh vs. Haryana

A comprehensive report on AQI levels, comparing two Indian states using average data from 2023 to 2024, visualized with an art-inspired theme.

Problem Statement

Data Handling and Comparison using AQI Data

Objective: To understand and compare the Air Quality Index (AQI) levels of two Indian states – Madhya Pradesh and Haryana – using real-life data through mathematical tools like averages, bar graphs, and percentage comparison.

Instructions:

- Collect Data: Use the table below which shows AQI values of five major cities each in Madhya Pradesh and Haryana. The data is from a given week. To be solved on A chart paper or A3 sheet . Decorate with art form of Madhya Pradesh .
- City (MP) AQI: Bhopal, Indore, Gwalior, Jabalpur, Ujjain
- City (Haryana) AQI: Gurugram, Faridabad, Panipat, Ambala, Rohtak

Tasks:

1. Calculate the average AQI for both Madhya Pradesh and Haryana. (Show all steps clearly.)
2. Make a bar graph comparing the AQI of each city. Use two different colors – one for MP cities and another for Haryana cities.
3. Determine which state has better air quality on average.
4. Write a few lines explaining your reasoning.

5. Find the difference between:
   The highest and lowest AQI among all 10 cities.

   The average AQIs of the two states.

6. Percentage Comparison: By what percentage is Haryana's average AQI higher than Madhya Pradesh's?

7. If the safe AQI level is considered to be 100, calculate how much each city exceeds the safe level and find the total excess for each state.

1. Objective and Data Overview

This analysis aims to understand and compare the Air Quality Index (AQI) levels of Madhya Pradesh and Haryana. We use average annual AQI data for five major cities in each state for the period of 2023 to 2024. The insights are presented through mathematical calculations, visual graphs, and interpretive explanations.

1.1. Collected AQI Data (Annual Average 2023-2024)

The following AQI values are the annual averages for the years 2023 and 2024, representing the overall air quality for each city during this period. Values are rounded for clarity.

Madhya Pradesh (MP) Cities & AQI

- Bhopal: **100**
- Indore: **101**
- Gwalior: **110**
- Jabalpur: **95**
- Ujjain: **98**

Haryana Cities & AQI

- Gurugram: **151**
- Faridabad: **157**
- Panipat: **125**
- Ambala: **117**
- Rohtak: **145**

2. Average AQI Calculation

The average AQI for each state is calculated by summing the AQI values of its cities and dividing by the number of cities. We show all steps clearly.

2.1. Madhya Pradesh Average AQI

To calculate the average AQI for Madhya Pradesh, we follow these steps:

1. **List AQI values for MP cities:** 100, 101, 110, 95, 98
2. **Sum of all AQI values:**

$$100 + 101 + 110 + 95 + 98 = 504$$

3. **Count the number of cities:** There are 5 cities in the list.
4. **Divide the sum by the number of cities to find the average:**

$$504 / 5 = 100.8$$

Madhya Pradesh Average AQI: 100.8

2.2. Haryana Average AQI

To calculate the average AQI for Haryana, we follow these steps:

1. **List AQI values for Haryana cities:** 151, 157, 125, 117, 145
2. **Sum of all AQI values:**

$$151 + 157 + 125 + 117 + 145 = 695$$

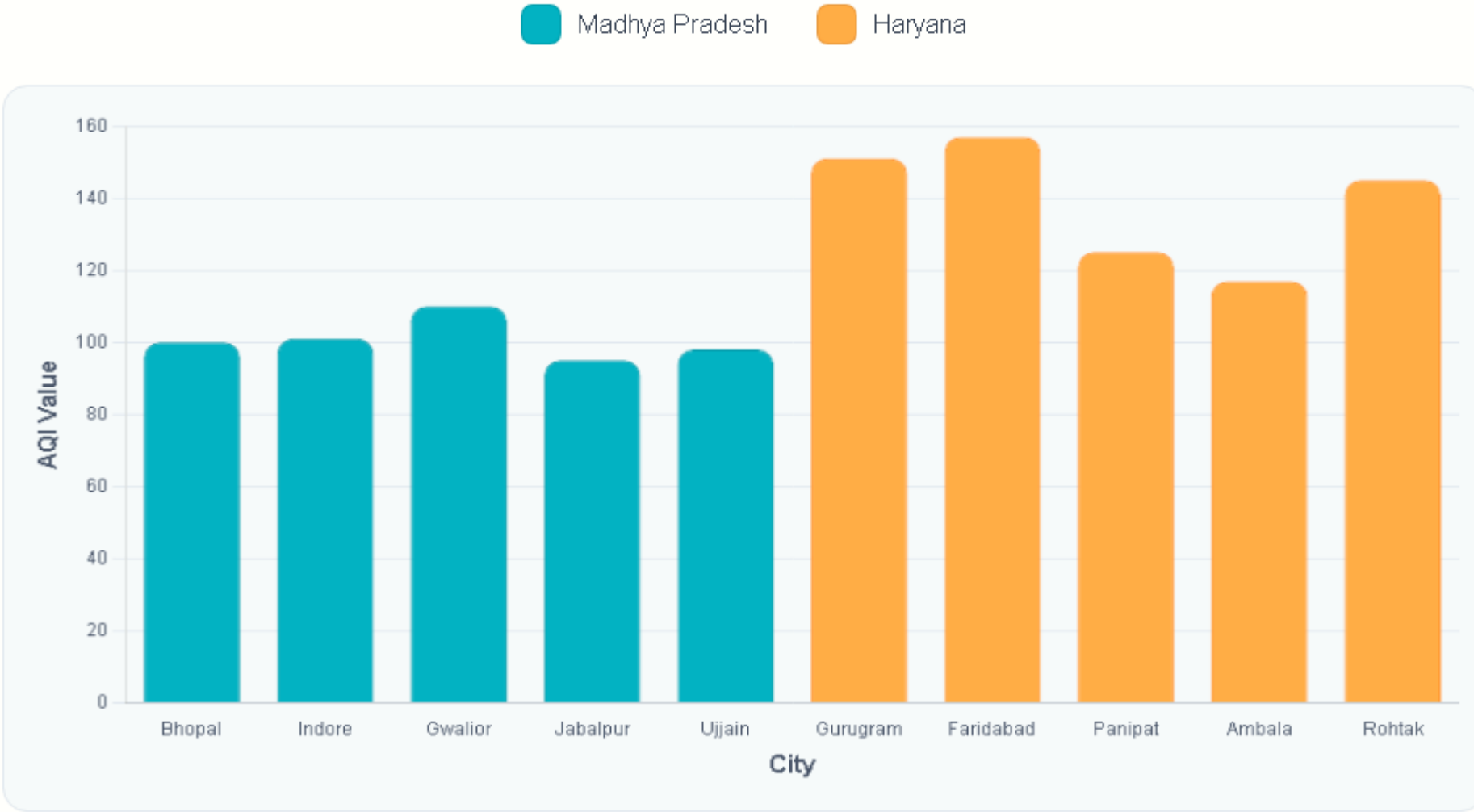
3. **Count the number of cities:** There are 5 cities in the list.
4. **Divide the sum by the number of cities to find the average:**

$$695 / 5 = 139$$

Haryana Average AQI: 139

3. AQI Comparison: Bar Graph

The bar graph below visually compares the average annual AQI (2023-2024) of each city. Madhya Pradesh cities are represented in **teal**, and Haryana cities in **orange**, reflecting a palette inspired by traditional Indian art forms.



4. Which State Has Better Air Quality?

Based on the average annual AQI values (2023-2024):

- Madhya Pradesh Average AQI: 100.8
- Haryana Average AQI: 139

Madhya Pradesh has better air quality on average.

Reasoning: A lower AQI value signifies superior air quality. Madhya Pradesh's average AQI of 100.8 is notably lower than Haryana's 139, indicating that the air quality in the selected cities of Madhya Pradesh is generally better than that in Haryana during the 2023-2024 period. This comparison highlights a significant difference in environmental conditions between the two regions.

5. Analysis of AQI Differences

5.1. Difference between Highest and Lowest AQI (All 10 Cities)

To find the difference between the highest and lowest AQI among all 10 cities, we follow these steps:

1. **List all 10 cities' AQI values:** [100, 101, 110, 95, 98, 151, 157, 125, 117, 145]
2. **Identify the Highest AQI:** 157 (Faridabad)
3. **Identify the Lowest AQI:** 95 (Jabalpur)
4. **Calculate the difference:**

$$157 - 95 = 62$$

Difference: 62

5.2. Difference between Average AQIs of the Two States

To find the difference between the average AQIs of the two states, we follow these steps:

1. **Haryana Average AQI:** 139
2. **Madhya Pradesh Average AQI:** 100.8
3. **Calculate the difference:**

$$\$139 - 100.8 = 38.2\$$$

Difference: 38.2

6. Percentage Comparison: Haryana vs. Madhya Pradesh

Here is the percentage by which Haryana's average AQI is higher than Madhya Pradesh's:

Haryana's average AQI is approximately 37.90% higher than Madhya Pradesh's.

7. Excess from Safe AQI Level (100)

Considering a safe AQI level to be 100, we calculate how much each city exceeds this level and the total excess for each state. This highlights the magnitude of pollution above desirable limits.

7.1. Madhya Pradesh Cities' Excess

To calculate the excess for Madhya Pradesh cities, we subtract the safe AQI level (100) from each city's AQI. If the result is negative (AQI below 100), the excess is considered 0:

- Bhopal: $100 - 100 = 0$
- Indore: $101 - 100 = 1$
- Gwalior: $110 - 100 = 10$
- Jabalpur: $95 - 100 = \text{0 (AQI is below safe level)}$
- Ujjain: $98 - 100 = \text{0 (AQI is below safe level)}$

Total Excess for Madhya Pradesh:

$$\mathbf{0 + 1 + 10 + 0 + 0 = 11}$$

Total Excess for Madhya Pradesh: 11

7.2. Haryana Cities' Excess

To calculate the excess for Haryana cities, we subtract the safe AQI level (100) from each city's AQI:

- Gurugram: $151 - 100 = 51$
- Faridabad: $157 - 100 = 57$
- Panipat: $125 - 100 = 25$
- Ambala: $117 - 100 = 17$
- Rohtak: $145 - 100 = 45$

Total Excess for Haryana:

$$\$51 + 57 + 25 + 17 + 45 = 195\$$$

Total Excess for Haryana: 195